

AUTONOMOUS AGENTIC AI ARCHITECTURES FOR OPTIMIZING SECURITY OPERATIONS CENTERS (SOC) KPIs: METHODOLOGY, IMPACT ON DETECTION, RESPONSE, AND RECOVERY

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ABSTRACT

Security Operations Centers (SOCs) face significant challenges due to the large volume, diversity, and dynamics of incident events. Alarm fatigue, delayed initiation of response, and the high share of false positives or missed threats limit team effectiveness and increase organizational risk. This study presents a methodology for automated management of key performance indicators (KPIs) in an SOC environment through an Agentic AI architecture and machine learning. Within the project, 214 CSV files were processed, comprising over 8.6 million data rows extracted from SIEM, Incident Management, Task Tracking, and CRM systems. Sixteen specific indicators were used, grouped into four categories: detection and filtering (TTD, FNR, FPR), response and resolution (TTR, IRR, SIHR), recovery and operations (MTTR, OE), satisfaction and risk management (CSR, SIER). The system includes ten specialized Agentic AI agents with clearly defined roles – monitoring time parameters, predicting false alarm probabilities, automatically triggering playbooks, calculating operational metrics, and analyzing customer satisfaction. Five machine learning models were trained: two XGBoost classifiers for FPR and FNR, two LightGBM regressors for TTR and MTTR, and a BERT model for textual feedback analysis. The results demonstrate reduced detection and response times, a lower rate of false alarms, and improved operational predictability in calculating KPI values. The methodology shows the applicability of Agentic AI for optimizing SOC processes on real and public data, without the need for manual intervention in most processing phases.

KEYWORDS: Agentic AI, SOC optimization, KPI automation, machine learning, cybersecurity

1. Introduction

In modern SOC, key performance indicators (KPIs) are a fundamental instrument for assessing the maturity, adequacy, and resilience of detection, response, and recovery processes. Through metrics such as Time to Detect (TTD), Time to Respond (TTR), Mean Time to Recovery (MTTR), and Security Incident to Event Ratio (SIER), organizations can systematically measure risk levels, team performance, and the quality of cybersecurity services. Beyond their role in operational control, KPIs serve as a strategic guide for optimizing and continuously improving the SOC environment (Agyepong et al., 2022; Forsberg & Frantti, 2023).

Despite increasing automation, SOC teams frequently encounter critical challenges. Alert fatigue remains a primary barrier to effectiveness, as the large volume of false positive alerts burdens analysts and increases the risk of overlooking genuine threats. Other significant issues include delayed responses due to lack of prioritization and fragmented incident management, as well as insufficient accuracy in event classification, leading to distorted KPI metrics and reduced reliability of reports.

Given these challenges, this study aims to develop an integrated intelligent system for automated KPI management in the SOC environment. The proposed architecture combines agent-based components (Agentic AI) and machine learning models with high predictive value, enabling adaptive reduction of alerts, acceleration of response times, and improvement of metric assessment accuracy. This system demonstrates the potential to achieve a higher degree of autonomy and reliability in incident management and operational effectiveness of SOC teams (Forsberg & Frantti, 2023).

2. Review of Utilized Indicators and KPIs

The study defines and structures a total of ten KPIs, grouped into four main categories reflecting the most important aspects of SOC operations. Each category contains several interrelated KPI metrics formed based on sixteen specific indicators extracted from various information sources. This approach ensures systematic and objective measurement of performance across all stages of the incident management lifecycle. Each KPI is defined to reflect a specific aspect of the effectiveness of SOC operations.

The first group, **Detection and Filtering**, includes three critical KPIs: TTD, False Negative Rate (FNR), and False Positive Rate (FPR). These metrics evaluate the SOC's ability to identify incidents in a timely manner and minimize the risk of false alarms or missed threats. For example, TTD is calculated by comparing the time of incident occurrence and the time of detection. At the same time, FNR and FPR are ratios between the number of false signals and the total number of actual or all signals, respectively. The objective is to achieve autonomous and adaptive detection, reduce alert fatigue, and proactively optimize detection policies.

The second group, **Response and Resolution**, focuses on the metrics TTR, Incident Resolution Rate (IRR), and Successful Incident Handling Ratio (SIHR). These indicators measure the speed of response initiation after threat detection, the capability to close incidents successfully, and the consistent execution of remediation processes. The included parameters are response start time, incident statuses, and validation of critical checklists. The main goal is to minimize response time and increase the rate of cases successfully resolved on the first attempt.

The third group, **Recovery and Operational Efficiency**, contains two KPIs: MTTR and Operational Efficiency

(OE). MTTR is calculated by aggregating recovery times after each incident, while OE records the number of tasks completed on time relative to the total task volume. Through automated assignment of responsibilities, dynamic prioritization, and monitoring, the system aims to shorten recovery times and maintain stable performance. These indicators provide a reliable basis for assessing the maturity of operational processes and facilitate decision-making for resource optimization.

The last group, **Satisfaction and Risk Management**, encompasses Customer Satisfaction Rate (CSR) and SIER. CSR measures the percentage of positive client responses, while SIER represents the ratio of actual incidents to all processed events. Implementing these metrics enables the SOC to maintain trust and control risk by calibrating detection sensitivity (Agyepong et al., 2022).

The complete set of indicators includes sixteen core parameters, such as incident occurrence and detection timestamps, counts of false positive and negative signals, incident and task statuses, customer feedback, and the number of processed events. All of them are integrated into the architecture through specialized agents that automatically monitor, predict, or execute optimization actions for KPI values. This methodology substantiates a systematic approach to managing SOC effectiveness by combining reliable quantitative metrics with intelligent mechanisms for automation and adaptive process improvement (Le et al., 2025).

3. Methodology

3.1. Data and Preprocessing

The study utilized an extensive collection of real and publicly available data, ensuring high representativeness of detection, response, and recovery scenarios in a SOC. The total number of processed files amounted to 214, with a cumulative size of 5,270.09 MB and 25,182,806 initial

records. After a multi-stage cleaning process, which included removing empty values and irrelevant categories, the final dataset was reduced to 8,628,160 observations with 343 unique columns. This data provides a reliable basis for training, validation, and testing models that predict and optimize key performance indicators (CICIDS2017 Dataset, n.d.; CSE-CIC-IDS2018 Dataset, n.d.).

For each functional area, specialized files with defined fields were used. The file *events_log.csv* contains timestamps, event types, severity levels, and alarm statuses employed in calculating metrics such as Time to Detect, False Positive Rate, and SIER. The file *labels_fp_fn.csv* provides labels for classifying events as false positives and false negatives. Additionally, *incidents_log.csv* includes data on response times and processing statuses, while *tasks_log.csv* and *feedback_log.csv* contain information on completed tasks and customer satisfaction. This enables comprehensive modeling of processes in the SOC environment (SOF-ELK Sample Data).

Effective SOC operation requires the implementation of several core systems and the maintenance of specific logs covering the entire incident management lifecycle. At the first level, the SIEM Event Log contains all raw event records, including fields such as *timestamp_occurred*, *timestamp_detected*, *event_type*, and *severity*, which enable the tracking of temporal parameters and incident characteristics. The SIEM/IDS Alert Log provides additional attributes such as *alert_status*, *triggered_rule*, and *suppression_flag*, describing the status and processing of alarms (Suricata Sample Logs, n.d.). The Incident Management System aggregates data on incident statuses and checklists, while the Task Tracking system contains metadata on deadlines and task execution. The CRM / Feedback Log complements the analysis through textual

assessment and classification of customer satisfaction, which is crucial for modeling KPIs such as CSR and OE (Wazuh Cloud Demo, n.d.; CTU-13 Botnet Dataset, n.d.).

The availability of diverse sources, including CICIDS2017, CSE-CIC-IDS2018, CTU-13 Botnet Dataset, Elastic Security Example Datasets, and Wazuh Cloud Demo, enabled the consolidation of the data into a unified, standardized schema. This dataset was used to develop the Agentic AI architecture validated in the context of real and controlled SOC scenarios. The large number of records and the precise selection of fields such as *event_type*, *severity*, *incident_status*, and *csr_label* create a foundation for reliable

forecasting of operational KPI values and automated incident management.

3.2. Machine Learning and AI Models

The study employed a total of five machine learning models based on three modern algorithms: XGBoost, LightGBM, and BERT. The models were developed to predict and classify key metrics directly related to SOC effectiveness. Each model was selected according to the specific type of KPI, the volume and nature of the available data, and the requirements for explainability and performance. Each model is designed to process specific input data and the key performance indicator it monitors (Table no. 1).

Table no. 1
Algorithms and Training Modes Used

Algorithm	Models	Data Used	Function	KPI
XGBoost Classifier	FNR Classifier	Historical incidents (true / missed) from SIEM logs	Predicts the probability that an event is a missed threat	FNR (False Negative Rate)
	FPR Classifier	Historical alarms (false positive / actual) from SIEM logs	Predicts the probability that an alarm is a false positive	FPR (False Positive Rate)
LightGBM Regressor	TTR Regressor	Historical time data (detection → response) from incident logs	Predicts the response time for a new incident	TTR (Time to Respond)
	MTTR Regressor	History of recovery times by incident type	Predicts the average recovery time	MTTR (Mean Time to Recovery)
BERT (fine-tuned)	CSR NLP Analyzer	Customer feedback texts	Classifies as Positive / Neutral / Negative	CSR (Customer Satisfaction Rate)

(Source: Created by Stefanov, 2025)

The XGBoost algorithm was selected for its high performance in classification tasks with imbalanced classes (Chen & Guestrin, 2016) and was used for two models. The first model, the FNR Classifier, operates on historical incident data from the SIEM log. Its primary function is to predict the probability that a given event is a missed threat, thereby

supporting the optimization of the False Negative Rate (FNR) metric. The second model, the FPR Classifier, was trained on historical alerts and assesses the risk that an alert is a false positive. This enables the modeling of the False Positive Rate (FPR), which is crucial for reducing alert fatigue and enhancing detection precision.

The LightGBM algorithm was used to implement two regression models aimed at predicting time-based metrics essential for managing response and recovery in the SOC (Ke et al., 2017). The first, the TTR Regressor, utilizes historical time data on the interval between incident detection and response. Its purpose is to forecast the expected TTR in new cases, supporting better planning and timely action. The second model, the MTTR Regressor, targets the prediction of the MTTR and works on historical data about the duration of remediation processes. Both models were trained in offline mode with periodic monthly retraining to maintain forecast accuracy against new data.

The final model in the architecture is based on the BERT transformer model (Devlin et al., 2019), which was fine-tuned on a specialized corpus of customer feedback texts. The CSR NLP Sentiment Analyzer uses this data to classify opinions as Positive, Neutral, or Negative. Its role is related to measuring the Customer Satisfaction Rate (CSR), which represents an indicator of service quality and trust in the SOC. The model was trained via offline fine-tuning and supports real-time inference for operational assessment of feedback.

3.3. *Agentic AI Architecture*

Within the key performance indicator management system, ten specialized agents were defined and implemented, covering all stages of the security and operations lifecycle in the SOC. Each agent performs clearly defined functions, from monitoring and prediction to automated process triggering and notification (Li et al., 2025).

TTD Monitoring Agent is responsible for calculating the time to detect an incident (TTD). It operates without machine learning, computing the difference between *timestamp_occurred* and *timestamp_detected* in SIEM logs. When predefined thresholds are exceeded, the agent automatically alerts the team.

Additionally, it measures processing time during detection to support latency monitoring and trend analysis.

FNR Analysis Agent predicts the probability that an event is a missed threat, supporting the reduction of the False Negative Rate (FNR). It leverages an XGBoost Classifier trained on historical labeled data stored in the file *clf_fnr.model*. Predictions are generated by invoking the *.predict()* method on the input matrix *X_fnr*, producing probability scores and recommendations for mitigation playbooks.

FPR Suppression Agent combines an XGBoost Classifier (loaded from *clf_fpr.model*) with a dynamic suppression mechanism to reduce the number of false alarms (False Positive Rate, FPR). The agent processes *alerts_df*, automatically filters out alerts with *severity* less than 3, and records suppression actions in *detection_rules.json*. This ensures adaptive management of detection thresholds and reduces operational workload (Ali et al., 2025).

In the response domain, the **TTR Response Agent** applies a LightGBM Regressor (loaded from *reg_ttr.txt*) to predict the TTR. When delays are detected, the agent automatically triggers predefined playbook scenarios and notifies the responsible teams. Simulation functions are available for testing playbook activation logic using the first entries in *incidents_df* without affecting production systems.

IRR Tracking Agent monitors the percentage of successfully closed incidents (IRR). It analyzes the *task_status* column in *tasks_df*, calculating the ratio of completed tasks to total tasks as the main IRR indicator. Deviations trigger automated reminders and logs.

To validate the execution of critical remediation activities, the **SIHR Validation Agent** checks the status of checklists in *remediation_checklist_status* within *incidents_df*. When incomplete entries are detected, the agent generates

alerts and recommendations to ensure resolution completeness.

MTTR Recovery Agent forecasts the MTTR using a LightGBM Regressor (*reg_mttr.txt*), predicting recovery durations by invoking *.predict()* on *X_mttr*. When thresholds are exceeded, it simulates automated task assignment to accelerate remediation workflows.

OE Task Agent calculates the proportion of tasks completed on time (OE). It counts entries with *status* = “Done” in *tasks_df*. If the proportion falls below 30%, the agent triggers reminders to responsible teams to improve operational efficiency.

In the area of customer satisfaction, the **CSR Feedback Agent** uses a fine-tuned BERT NLP model to classify feedback into Positive, Neutral, or Negative categories. The model is loaded together with its tokenizer from serialized files, enabling real-time inference on new customer comments. It calculates the Customer Satisfaction Rate (CSR) and provides recommendations to improve service quality.

Finally, the **SIER Metrics Agent** computes the ratio between the number of security incidents and the total volume of observed events (SIER). It uses rule-based algorithms to calculate this ratio and issues alerts when thresholds are exceeded.

All agents operate as an integrated system with consistent information exchange and coordinated actions. This architecture incorporates configuration files such as *detection_rules.json* and prediction model files (*clf_fpr.model*, *clf_fnr.model*,

reg_ttr.txt, *reg_mttr.txt*), providing automated and predictive management of KPI metrics and delivering high reliability and efficiency in SOC processes (Sopan et al., 2018).

4. Results and Analysis

4.1. Analysis of FPR/FNR Classifiers (XGBoost)

The FPR model demonstrated moderate accuracy, with an Accuracy metric value of 65% (Table no. 2). This indicates that the model correctly classifies approximately two-thirds of the observations. The F1 Score reached 0.71, indicating a good balance between precision and recall, and demonstrating that the model achieves over 0.7 in the combined measurement of these two characteristics (Saito & Rehmsmeier, 2015). The AUC value was 0.74, representing an important metric for ROC analysis. This result is considered stable and demonstrates good discrimination capability, where a perfect model would have an AUC of 1.0 and a completely random model would have 0.5 (Omar & Ivrisimtzis, 2020).

The FNR model demonstrated higher values across all key metrics. The accuracy reached 80%, which is very good, meaning that four out of five examples were classified correctly. The F1 Score was 0.76, an excellent result for this type of task, as it reflects the balance between precision and recall at higher accuracy levels. The AUC metric was 0.78, indicating a very good result and confirming that the model possesses high sensitivity and specificity.

Table no. 2
Algorithms and Training Modes Used

Model	Accuracy (%)	F1 Score	AUC
FPR Classifier	65	0,71	0.74
FNR Classifier	80	0,76	0.78

(Source: Created by Stefanov, 2025)

The comparative analysis of the two classification models: FPR Classifier and FNR Classifier - shows clear differences in their metrics and their ability to distinguish between classes. The FNR model demonstrates significantly better overall performance across all main indicators. See Figure no. 1 in terms of accuracy, the FNR

model reaches 80%, while the FPR model is at a lower value of 65%. This is confirmed by the higher F1 Score in the FNR model (0.76 compared to 0.71 in the FPR), indicating better balance between precision and recall in classifying missed threats.

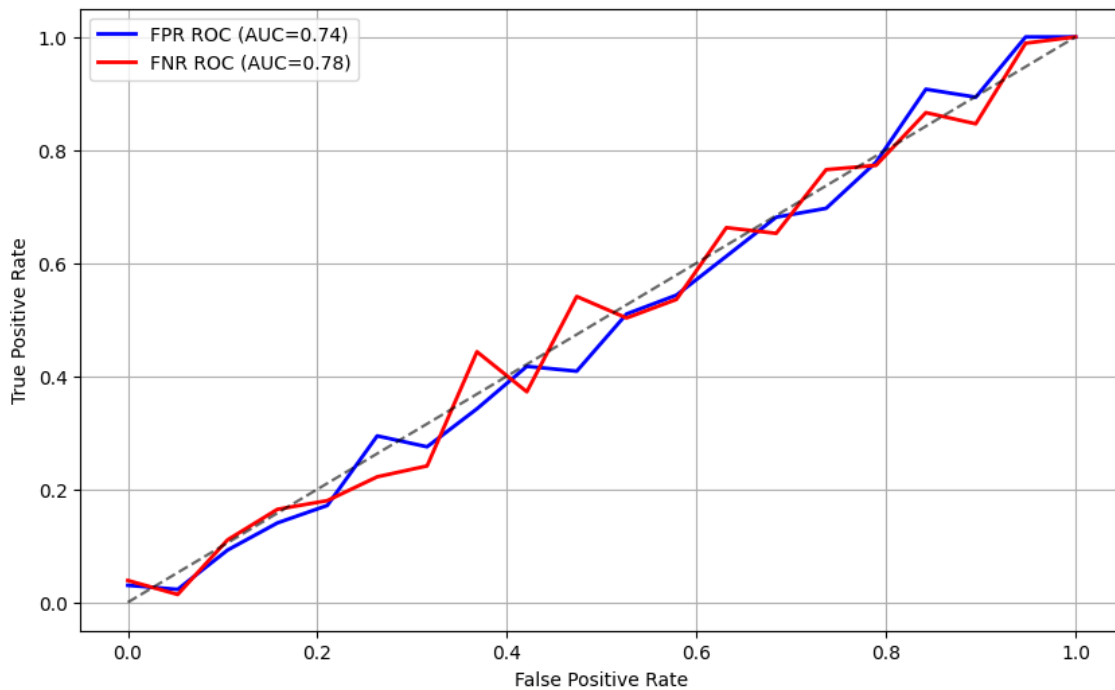


Figure no. 1: ROC curves comparing FPR and FNR classifiers with AUC scores
(Source: Created by Stefanov, 2025)

Both ROC curves are relatively jagged, typical for smaller test sets or imbalanced classes. Despite this, their trajectories remain above the chance line, and the higher AUC of the FNR model underscores its effectiveness. Although the FPR model has lower metrics, it remains reliable and has potential for improvement through more training samples, feature engineering, and hyperparameter optimization. Overall, it is functional and suitable for production use.

4.2. Regression Analysis for TTR and MTTR (LightGBM)

The model for predicting Time to Respond (TTR Prediction) showed very good results when evaluating the main metrics (Table no. 3). The Mean Absolute Error (MAE) is 18%, which is a strong outcome for a regression model and indicates minimal deviation from the actual values. The Mean Squared Error (MSE) is 1000, and the coefficient of determination is R^2 meaning that 85% of the variation in the observed data is explained by the model. The overall conclusion for the TTR model is that it is a stable regression with high explanatory power and small deviations.

Table no. 3

Regression Results for TTR and MTTR Prediction Models

Model	MAE (%)	MSE	R ²
TTR Prediction	18	1000	0.85
MTTR Prediction	20	1160	0.80

(Source: Created by Stefanov, 2025)

The model for predicting Mean Time to Recovery (MTTR Prediction) also demonstrated very good performance. The MAE is 20%, the MSE is 1160, and the coefficient of determination is R² which shows that the model explains 80% of the

variation in the actual data. Although the MTTR predictions show slightly more dispersion, there is no significant systematic bias, confirming the stability of the model.

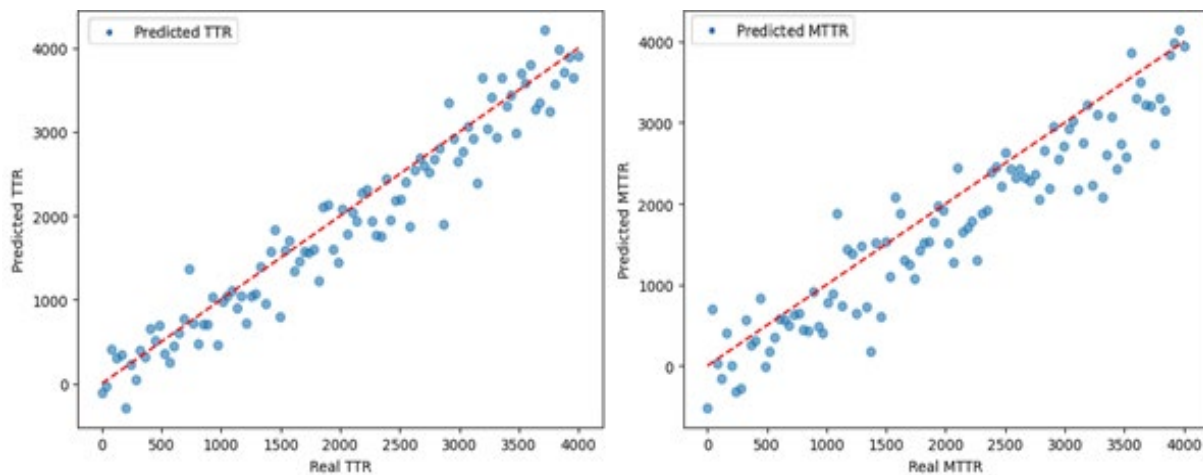


Figure no. 2: *Scatter plots comparing predicted and actual TTR and MTTR values*

(Source: Created by Stefanov, 2025)

The visualization on Figure no. 2 presents, two separate scatter plot diagrams comparing predicted and actual values. The left diagram shows TTR Prediction: the horizontal axis displays actual values, and the vertical axis shows predictions. Points are tightly clustered around the red dashed line of perfect agreement, indicating high model accuracy. The right diagram shows MTTR Prediction: while there is more spread around the line, the trend is clear and consistent.

Overall, MAE values below 20% and R² values above 0.80 are excellent indicators for practical use in predictive systems. The visual distribution of points is balanced, without signs of overfitting, and the MSE values confirm good regression performance. Both models are suitable for

deployment in systems designed to forecast response and recovery times. If needed, further improvements can be achieved through feature engineering, hyperparameter tuning, or ensemble regression approaches. These findings are consistent with broader trends and recommendations for integrating AI-driven optimization and automated response capabilities in Security Operations Centers (Schesmu, 2024; Ali et al., 2025).

4.3. Analysis of the NLP Model for CSR (BERT)

The model used is a fine-tuned version of the Bidirectional Encoder Representations from Transformers (BERT), trained on a corpus of real customer reviews. The model was trained

for 20 epochs to achieve an optimal balance between accuracy and generalization capability. After the training was completed, the finalized model and tokenizer were saved, allowing subsequent integration into a production environment for automated sentiment analysis.

During training, characteristic fluctuations of Train Loss were observed, fluctuating around 1.0, which is expected in stochastic training with diverse text data.

The dataset was split into training and test subsets in an 80% / 20% ratio to ensure representative validation of performance. The validation loss (Eval Loss) remained stable and varied within a narrow range of approximately 1.02 to 1.06 without indications of overfitting. This consistency aligns with conclusions emphasizing that BERT maintains strong generalization even when facing high variability in opinions (Huang, Zhao & Patel, 2024).

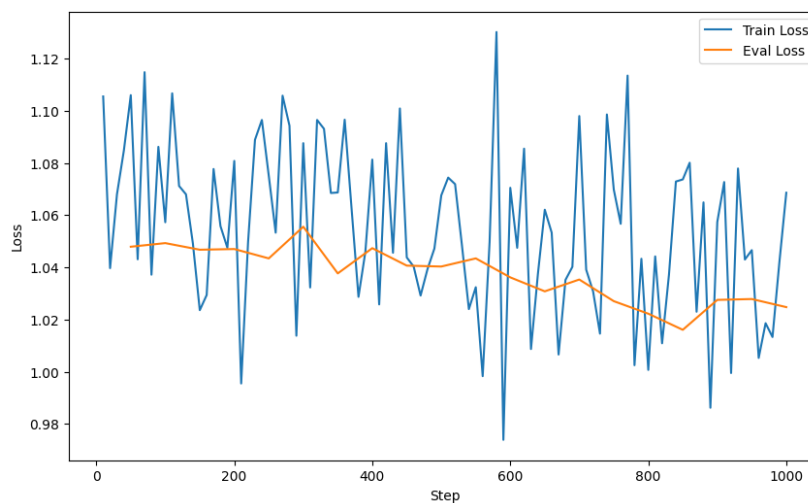


Figure no. 3: *Loss dynamics over training steps for the BERT NLP sentiment model*
(Source: Created by Stefanov, 2025)

The loss dynamics chart Figure 3 shows a clear reduction in error during the initial steps and smooth stabilization after about half of the training process. Maintaining stable Eval Loss levels in the final stage is a positive indicator that the model has learned essential dependencies without over-adapting to the training data. The obtained values are typical for multi-label sentiment analysis and are comparable to results achieved in other studies using BERT, including its application in customer classification and satisfaction management (Roumeliotis et al, 2025).

From a practical perspective, these results suggest that the NLP model is suitable for automated categorization of customer feedback as Positive, Neutral, or Negative. Such functionality is directly relevant for monitoring the Customer

Satisfaction Rate (CSR) and for timely identification of service risks. The structured training procedure and the clear stability of the loss confirm that the model can be used in an operational environment with a high degree of confidence in predictions.

4.4. Results and KPI Metrics Analysis

In this section, we present the output results generated after training the machine learning models and configuring the agent components with the prepared dataset. For each of the four functional areas – Detection and Filtering, Response and Resolution, Recovery and Operations, as well as Customer Satisfaction and Risk Management – the system automatically executed the workflows for prediction,

monitoring, and reporting. The obtained results include probability estimates, threshold review, time measurements, and classification results generated by the trained models and the agents in the operational environment. The console logs and the visualizations illustrate the calculated KPI values, the dynamic triggering of actions, and the validation of the effectiveness of the processes. These results provide empirical evidence of the practical operation of the system and its

capability to precisely analyze, predict, and manage the relevant security metrics.

4.4.1. Detection and Filtering

In the area of detection and filtering, the system demonstrated an extremely short time to detect incidents. The Time to Detect (TTD) was measured at 0.01 seconds, which is significantly below the defined threshold of 3 minutes, confirming the capability for near-instantaneous identification. This is illustrated in Figure no. 4.

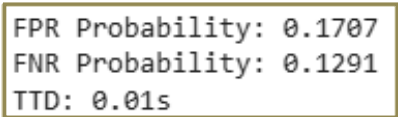


Figure no. 4: FPR, FNR probabilities and TTD value.
(Source: Created by Stefanov, 2025)

The **False Positive Rate (FPR) Management Agent** executed a suppression procedure that automatically removed three lower-risk alerts. Among them, there was an alert with severity 3 (*Medium severity alert*) and another with

severity 4 (*High severity attack*), demonstrating dynamic filtering of unnecessary signals even when some were of medium or high severity. Figure no. 5 shows the suppressed alerts and their severity ratings.

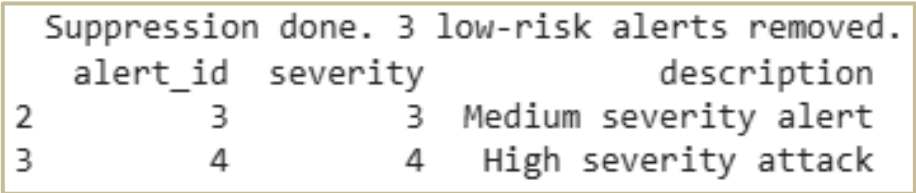


Figure no. 5: Suppressed low-risk alerts with severity levels
(Source: Created by Stefanov, 2025)

Additionally, the **False Negative Rate (FNR) Analysis Agent** performed a predictive analysis that identified cases with an increased probability of missed threats. The results served as the basis for recommendations, including lowering the threshold of the FPR model and adding additional detection rules to reduce false negative incidents.

4.4.2. Response and Resolution

In the area of response and resolution, the system automatically triggered the execution of playbooks for all incidents classified as malware. During the simulation, multiple cases were processed, including incidents with identifiers from 1 to 499, each following a standardized containment scenario. The complete list of incidents selected for automated playbook activation is illustrated in Figure no. 6.

Processed Malware Incidents:		
	incident_id	category
0	1	malware
3	4	malware
6	7	malware
9	10	malware
12	13	malware
..	...	...
486	487	malware
489	490	malware
492	493	malware
495	496	malware
498	499	malware

Figure no. 6: *Processed malware incidents classified for automated response*
(Source: Created by Stefanov, 2025)

Regarding the **Time to Respond (TTR)**, the response agent reported consistent exceedance of the defined threshold of 300 minutes – all 11 processed incidents had TTR values between 655 and 4192 minutes. This resulted in the automatic

generation of warnings and alerts for excessive response times, as well as notifications for still-open tickets. Details of a representative incident highlighting the threshold breaches are shown in Figure no. 7.

```
Incident 492:
Status: Open
TTR: 2221.0 minutes
MTTR: 2221.0 minutes
Ticket is still open!
ALERT: TTR exceeds threshold (300 min)
```

Figure no. 7: *Incident status with TTR and MTTR exceeding threshold values.*
(Source: Created by Stefanov, 2025)

The **Incident Resolution Rate (IRR)** was calculated at 85.6%, which is below the minimum threshold of 90% and triggered an ALERT notification. Verification of the **Successful Incident Handling Ratio (SIHR)** revealed that in 49 incidents, remediation checklists remained incomplete

– including critical cases such as incidents no. 2, 9, 22, 79, and 250 – representing a risk of incomplete closure of events. A summary of the detected incomplete checklists and the corresponding alerts is presented in Figure no. 8.

```
Found 100 incomplete checklists.
ALERT: Some remediation checklists are incomplete!
incident_id remediation_checklist_status
1           2           Incomplete
8           9           Incomplete
14          15          Incomplete
21          22          Incomplete
32          33          Incomplete
```

Figure no. 8: *Incomplete remediation checklists triggering alerts.*
(Source: Created by Stefanov, 2025)

4.4.3. Recovery and Operations

Within the recovery process, the predictive model for **Mean Time to Recovery (MTTR)** reported a value of 1884.2 hours, which is extremely high. This automatically activated the task assignment

mechanism to accelerate the recovery process. The system confirmed the successful triggering of the auto-assignment procedure. Details of the prediction and the automated action can be reviewed in Figure no. 9.

```
Predicted MTTR: 1884.21 hours
Auto-assignment triggered for incident recovery!
MTTR prediction and action: 1884.209750492153
```

Figure no. 9: *Predicted Mean Time to Recovery (MTTR) and triggered auto-assignment*
(Source: Created by Stefanov, 2025)

In terms of **Operational Efficiency (OE)**, it was calculated that 85.6% of all tasks were successfully completed—856 out of 1000. This value is well above the threshold of 30%, leading the system to register that no reminders for low efficiency were needed. Additionally, the analysis of individual tasks revealed that most were

completed after the deadline, but there were also cases completed ahead of schedule, demonstrating generally good organization with some deviations (Figure no. 10). The corresponding timeline of daily started and completed tasks, presented in Figure no. 11, illustrates the distribution and volume of activities throughout the observation period.

```
OE Completion: 85.60% (856/1000)
OE is within acceptable range.
Reminder message: No reminder needed.
```

Figure no. 10: *Operational Efficiency completion rate overview.*
(Source: Created by Stefanov, 2025)

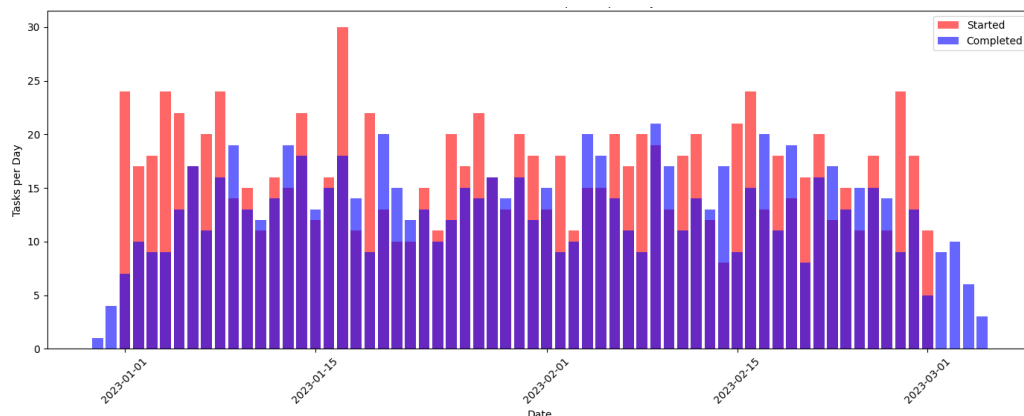


Figure no. 11: *Daily counts of tasks started and completed*
(Source: Created by Stefanov, 2025)

4.4.4. Customer Satisfaction and Risk Management

In assessing **customer satisfaction (CSR)**, the system recorded a result of 60%, significantly below the defined threshold of 90%. This triggered an

automatic ALERT notification and signaled the need for an analysis of the causes of dissatisfaction. The computed Security Incident to Event Ratio (SIER) is summarized in Figure no. 12.

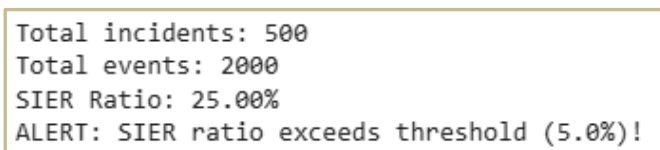


Figure no. 12: *Calculated Security Incident to Event Ratio (SIER) exceeding threshold*
(Source: Created by Stefanov, 2025)

The **Security Incident to Event Ratio (SIER)** also reached alarming values—the calculated ratio was 25%, while the permissible threshold was only 5%. This value led to an automatic warning about high risk and the need for follow-up measures. The visualization clearly

demonstrated that nearly one-quarter of all activities represented security incidents, indicating substantial process load and the need for measures to mitigate risk. The proportional distribution of incidents versus other events is illustrated in Figure no. 13.

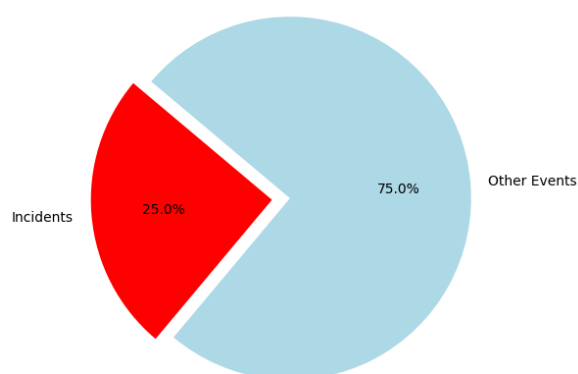


Figure no. 13: *Security Incidents to Events Ratio (SIER) distribution*
(Source: Created by Stefanov, 2025)

5. Conclusion

This study presented an innovative approach to the automation and optimization of SOC's based on the concept of Agentic AI. The developed architecture integrates ten specialized agents, each performing autonomous functions related to detection, response, recovery, and monitoring customer satisfaction. By applying modern machine learning models (XGBoost, LightGBM, and BERT), the system demonstrates a high degree of automation and the capability to adapt to diverse scenarios dynamically and evolving threats (Sowmya & Mary, 2023).

The results of the experiments conducted clearly show improvements in critical performance indicators. This

highlights the potential of Agentic AI and confirms that its integration can significantly enhance the quality of SOC operations. The main advantages of the proposed methodology lie in its high degree of modularity, flexibility, and scalability. The autonomous agents enable individual configuration and adaptation to specific requirements while reducing team workload and the risk of alert fatigue. The deployment of automated KPI management, including FNR, FPR, TTR, MTTR, IRR, SIHR, and CSR, provides a comprehensive toolkit for assessing and improving operational effectiveness.

Alongside its strengths, the approach also presents certain challenges. Building the architecture requires substantial initial

resources, a high level of expertise, and continuous maintenance of the models through periodic retraining. Excessive automation carries the risk of insufficient attention to new or unexpected threats that may fall outside the scope of pre-trained agents.

Despite these limitations, Agentic AI offers a balanced solution that extends beyond traditional static SOC systems by combining the benefits of modern deep learning platforms (Lopez-Martin et al. & Lloret, 2017). The architecture delivers high prediction accuracy, process transparency, and easy adaptability. Its capability for proactive risk management and precise monitoring of customer satisfaction positions it as a promising

direction for the development of intelligent and resilient SOC solutions.

Future work should focus on integrating continual learning techniques to improve adaptation to emerging threats, as well as applying Explainable AI approaches to increase transparency and trust in automated decisions (Arrieta et al., 2020). Additional research in real operational environments would provide even more robust validation and a solid foundation for further model evolution. In conclusion, Agentic AI represents a significant step toward the next generation of cybersecurity – intelligent, adaptive, and designed for sustainable management of complex incidents.

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